

Weather Trend Forecasting Report

This report provides a structured overview of the project in which a comprehensive data analysis is performed on a weather trends data comprising 2 years of data. The objective of this stage is to develop a forecasting model capable of predicting future temperature values using historical weather observations. The dataset contains multiple environmental variables, including humidity, atmospheric pressure, wind speed, cloud cover, UV index, geographical coordinates, and time-based information. Different machine learning approaches are implemented and compared to determine the most effective method for temperature prediction.

Data Cleaning

Initial data exploration shows that the dataset contains 41 columns and 155336 rows. No null or duplicate values are found in the dataset in the initial cleaning process. However, in the second phase of data cleaning, for one data entry, location name and last updated columns contained the same information twice with only difference in the sun and the moon rising and setting times. Hence, one of the two rows is removed. Some columns are removed from the original dataset because they are the duplicates of the same values but are using different measuring units for metrics. Such as `temperature_fahrenheit` represented the same temperature information as `temperature_celsius`, but in Fahrenheit. `Wind_mph` duplicated the wind speed measurements already recorded in kilometers per hour (`wind_kph`). `Pressure_in` measured atmospheric pressure in inches of mercury, while the same information is already available in millibars (`pressure_mb`). `Precip_in` is removed because precipitation is already recorded in millimetres (`precip_mm`). `Visibility_miles` is removed because visibility measurements in kilometres (`visibility_km`) contain the same information. `Gust_mph` is removed because wind gust speed is already available in kilometres per hour (`gust_kph`). `Feels_like_fahrenheit` is removed because it duplicated the apparent temperature information already recorded in Celsius (`feels_like_celsius`).

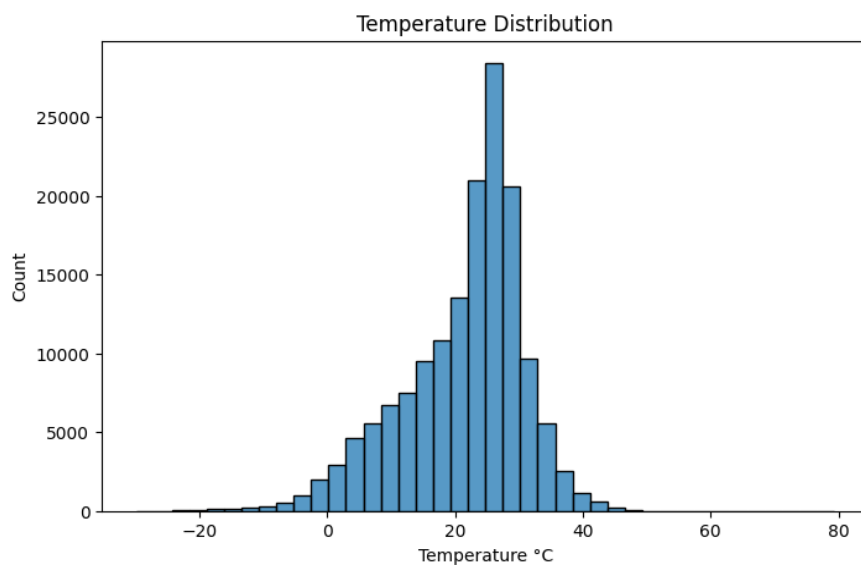
Validity Checks

Validity checks are performed on all numerical weather variables using scientifically accepted meteorological and environmental ranges. Air temperature is expected to fall between -50°C and 55°C , while apparent (feels-like) temperature is considered valid between -60°C and 60°C . Wind speed and wind gusts are checked against maximum realistic values of 200 km/h and 220 km/h, respectively. Atmospheric pressure is expected to lie between 850 mb and 1100 mb, covering the range of normal sea-level and extreme weather conditions. Air quality variables are also validated. PM2.5 values are inspected above $500\text{ }\mu\text{g}/\text{m}^3$, PM10 values above $1000\text{ }\mu\text{g}/\text{m}^3$, Carbon Monoxide (CO) above $10,000\text{ }\mu\text{g}/\text{m}^3$, and Sulphur Dioxide (SO_2) above $500\text{ }\mu\text{g}/\text{m}^3$ and UV Index between 0 and 20. Similar checks are performed for precipitation values and visibility. Records containing physically impossible values, such as extremely high temperatures or negative particulate matter concentrations, are treated

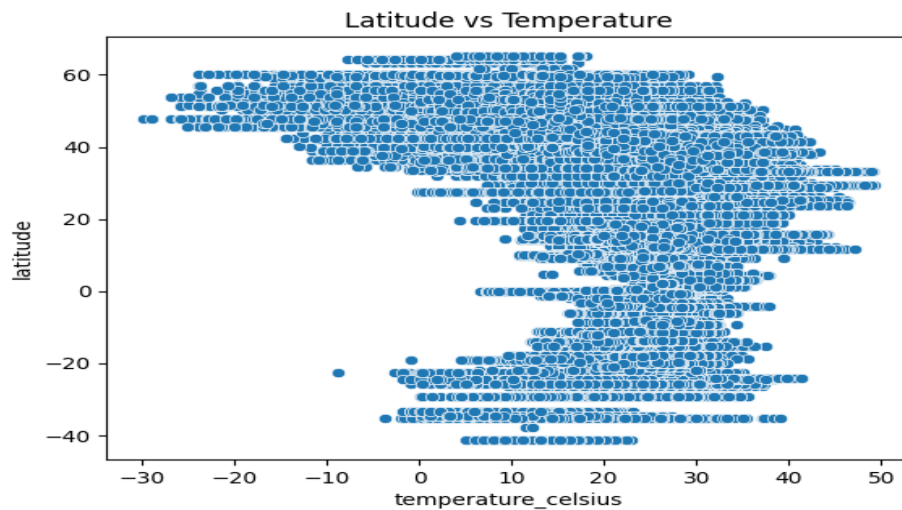
as data errors and replaced with missing values before imputation using the median for the corresponding location. However, unusually high pollution measurements that are consistent with known severe pollution events are retained, as they represent plausible environmental conditions rather than data entry errors. This validation process ensured that the dataset remained scientifically realistic while preserving genuine extreme weather and air quality observations. After cleaning, the dataset has 155334 rows and 34 columns.

Exploratory Data Analysis

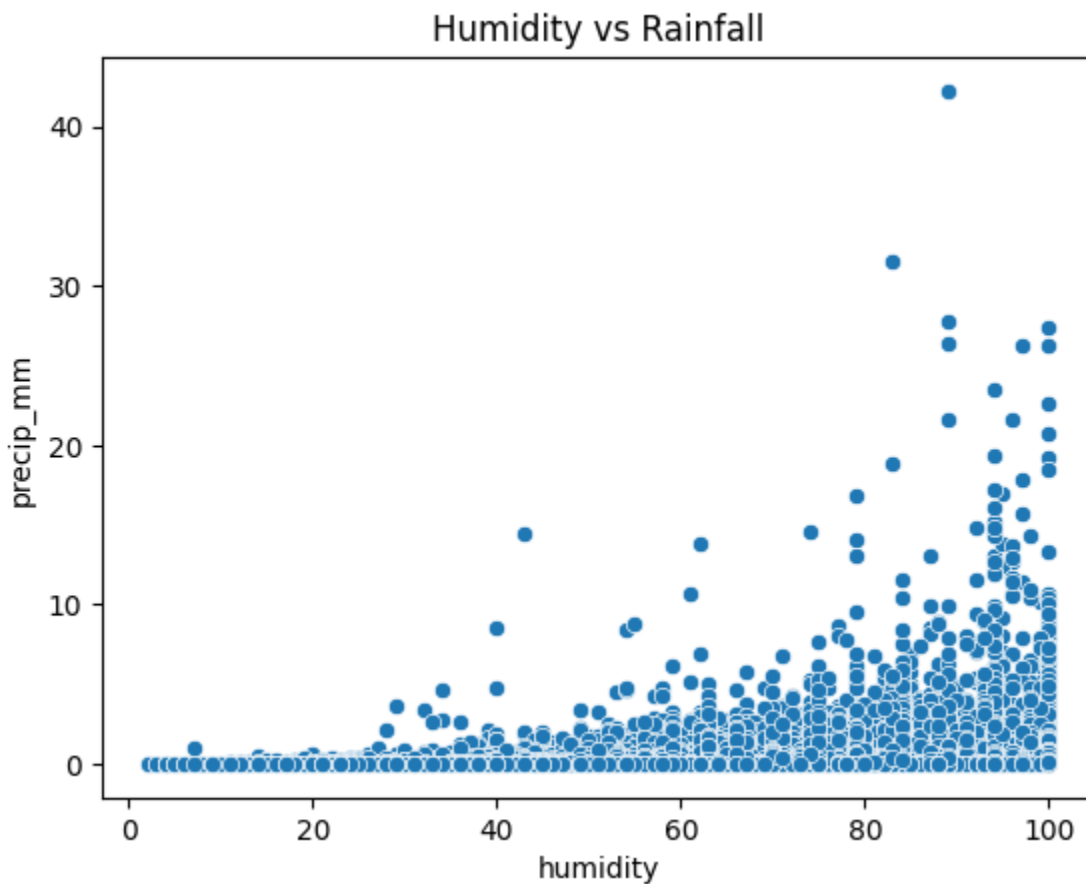
In exploratory data analysis, the number of unique countries and unique locations are identified which came out to be 211 and 268 respectively. For most countries, temperature is distributed between 0 to 40 Degrees Celsius as the graph shows. While the extreme temperatures are recorded at -29°C and 79°C , later of which is found to be unreasonable for the country and condition and considered a misreading and hence normalized using median.



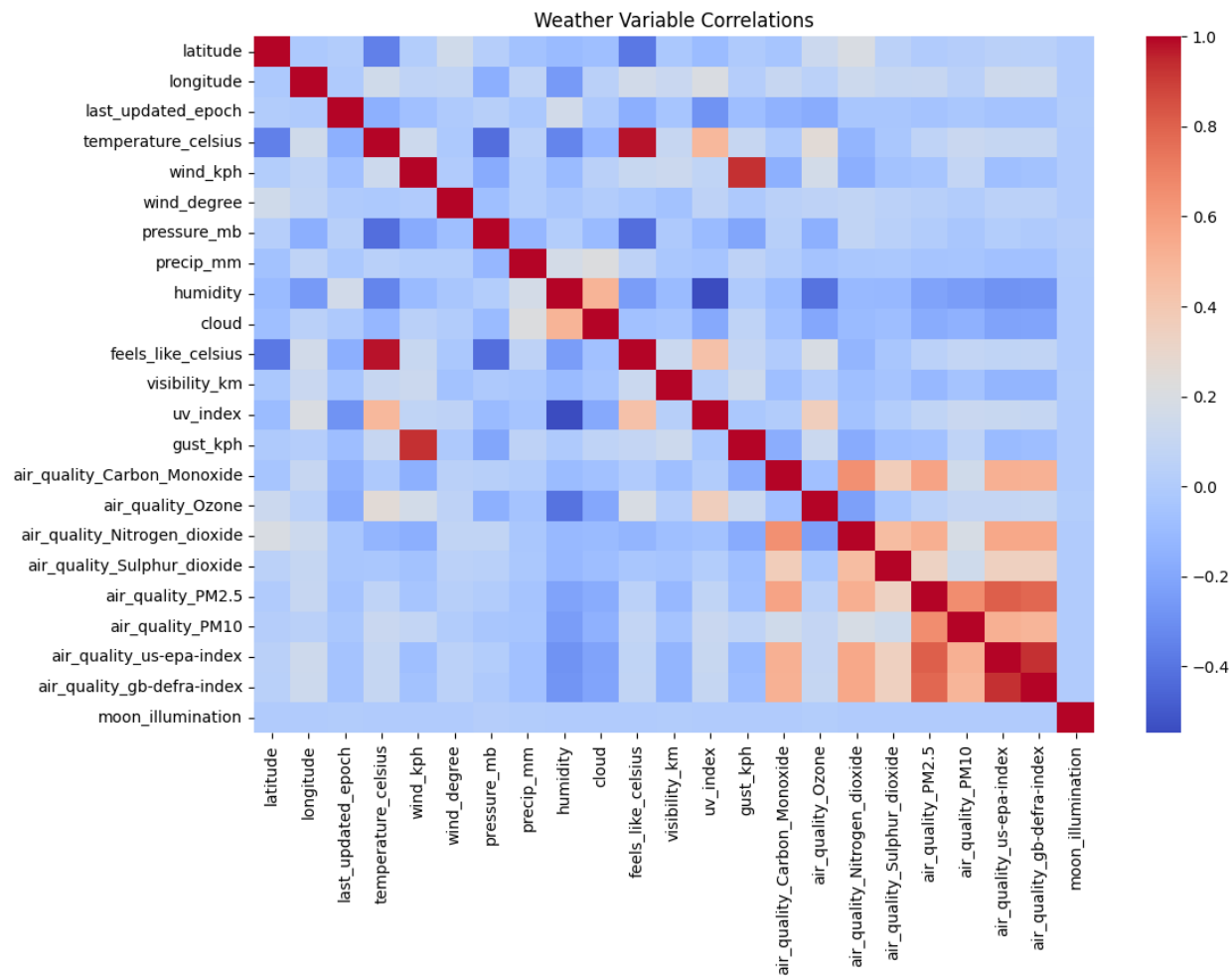
The hottest location found to be Ar Riyadh with 45°C temperature recorded. The relationship between latitude and temperature is observed in a scatterplot and it is found that the temperature in high latitudes inclines more towards negative while on the southern side, the temperature is usually rising and, on the equator, the temperature stays normal without fluctuating to extreme temperatures. The plot below shows this.



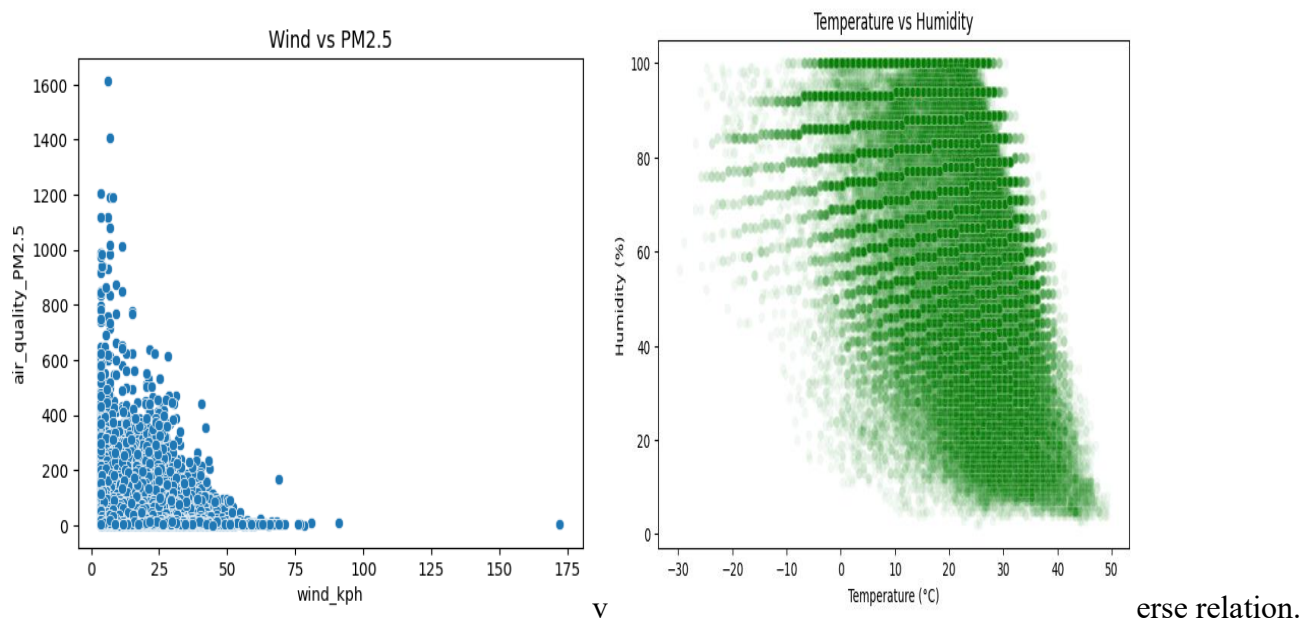
Precipitation and humidity also showed a positive relation where at low humidity level, precipitation and rain fall probability is low and at high humidity level, precipitation and rain fall propbality is higher. The plot below shows this.



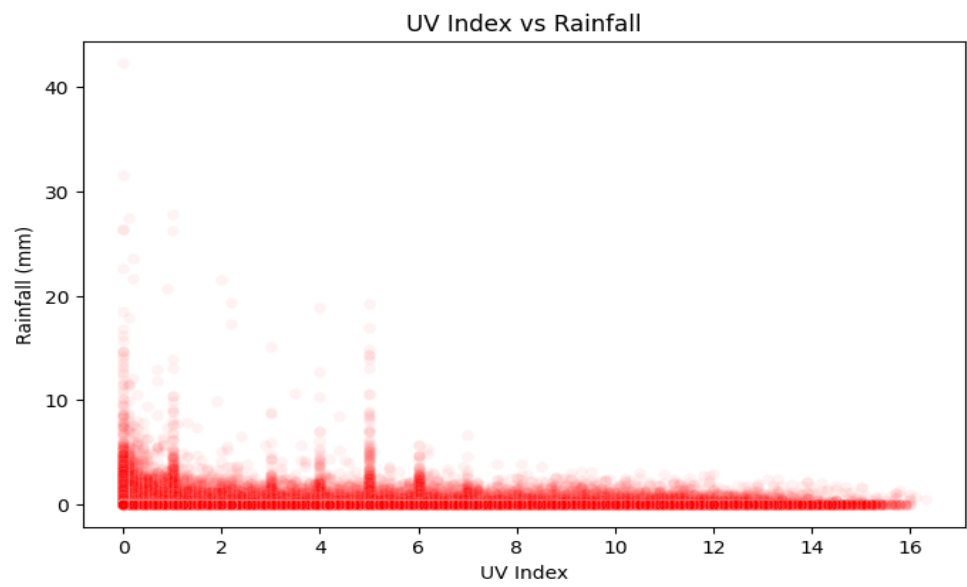
A heatmap also shows the patterns where the air quality indicators are found to have positive relation with each other, as the higher value of one also indicates towards the presence of high value of other air quality indicators. Heatmap also shows a positive relationship between cloud and humidity levels. This observation can be seen from the chart below.



Some other patterns found between weather metrics such as Wind and PM 2.5 are found to have an inverse relation; humidity and temperature also show inverse relation to some extent where temperature increases and humidity decreases.

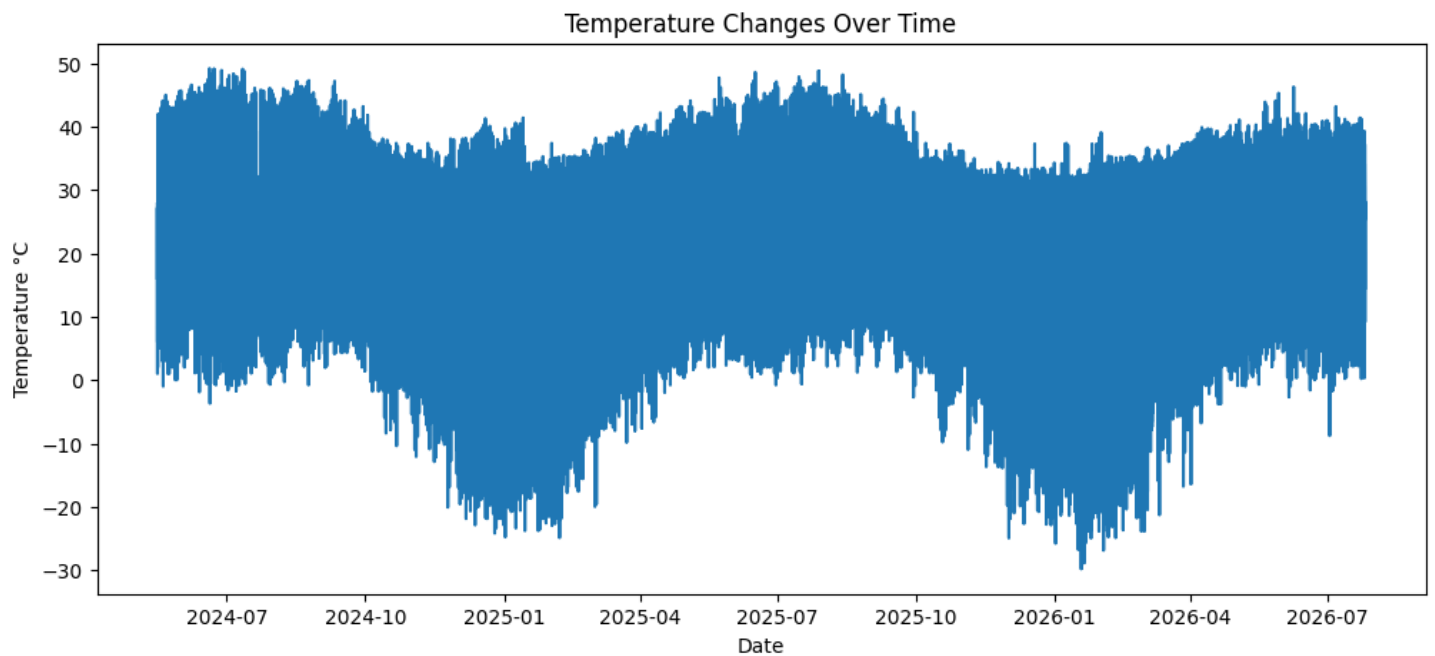


Similarly, the UV index and rainfall also show inverse relations.



Time Series Analysis

The time series analysis show the temperature changes over time between July 2024 to July 2026. Temperature is seen to be rising during summer overall, while it is seen to be dropping during winter months. Daily temperatures also showed similar trends except for a few outliers.



Data Modeling for Weather Forecasting

Before developing the machine learning models, a baseline forecasting model is created to provide a benchmark for comparison. The baseline model predicts the next temperature using the most recently observed temperature, based on the simple assumption that future temperature is likely to be like the current temperature. This approach serves as a reference point for evaluating whether more advanced machine learning models could achieve improved forecasting accuracy.

Data Preparation

The dataset is first sorted chronologically using the `last_updated` timestamp to preserve the temporal order of weather observations. To capture time-dependent weather patterns, two additional features are extracted from the timestamp: the hour of the day, representing daily temperature variations, and the month, representing seasonal changes. The machine learning models are trained using humidity, atmospheric pressure, wind speed, cloud cover, UV index, latitude, longitude, hour, and month as the predictor variables, while temperature in degrees Celsius (°C) is selected as the target variable for forecasting.

Train-Test Split

The dataset is divided into training and testing sets using a time-based split, ensuring that earlier observations are used for training and later observations are used for evaluation.

Three forecasting approaches are implemented.

Baseline Model

The baseline model predicts future temperatures using the previous temperature value. This provided a simple benchmark to measure whether machine learning models could improve forecasting accuracy.

Random Forest Regression

Random Forest uses multiple decision trees to identify relationships between weather variables and temperature. The model can learn complex relationships between Geographic location, Atmospheric conditions, Weather patterns and Seasonal effects.

XGBoost Regression

XGBoost uses sequential decision trees where each new tree attempts to correct previous errors. This allows the model to capture complex non-linear relationships within weather data.

Ensemble Model

An ensemble model is created by combining Base Model, Random Forest and XGBoost predictions. The final prediction is calculated by averaging predictions from all three models. The purpose of the ensemble was to reduce individual model weaknesses and improve overall forecasting performance.

Evaluation Metrics

The models is evaluated using three performance metrics:

Mean Absolute Error (MAE)

MAE measures the average difference between predicted and actual temperatures. A lower MAE indicates more accurate predictions

Root Mean Squared Error (RMSE)

RMSE measures prediction of errors while giving more importance to larger mistakes. A lower RMSE indicates fewer significant prediction errors.

R² Score

R² measures how much variation in temperature is explained by the model. Values closer to 1 indicate stronger predictive performance.

Model Performance Results

Model	MAE (°C)	RMSE (°C)	R ² Score
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Baseline (Previous Temperature)	7.72	10.44	-0.361
Random Forest	2.28	3.22	0.870
XGBoost	2.33	3.19	0.873
Ensemble Model	2.16	3.00	0.887

Discussion

The results show that temperature forecasting benefits from using multiple environmental variables rather than relying only on previous temperature measurements. The baseline model produced the weakest performance, with an MAE of 7.72°C and a negative R^2 score. This indicates that simply using the previous temperature value was not sufficient for accurate prediction.

Both Random Forest and XGBoost significantly improved forecasting performance. The models achieved R^2 values above 0.87, demonstrating that weather variables such as humidity, pressure, wind speed, geographical location, and time information provided valuable predictive information.

Random Forest achieved slightly lower MAE compared with XGBoost, while XGBoost achieved a slightly higher R^2 score and lower RMSE. This suggests that both models performed similarly but captured different aspects of temperature behaviour.

The ensemble model achieved the best overall results, reducing MAE to 2.16°C and increasing the R^2 score to 0.887. This demonstrates that combining multiple models improved prediction accuracy by reducing individual model errors.

The strong performance of machine learning models indicates that temperature patterns are influenced by several interacting factors, including:

- Geographic location
- Seasonal variation
- Atmospheric pressure
- Humidity
- Wind conditions
- Cloud coverage

The ensemble approach provided the most reliable predictions because it combined the strengths of both Random Forest and XGBoost.

Conclusion

This study successfully developed and evaluated several temperature forecasting models. The baseline model demonstrated limited forecasting ability, while machine learning approaches achieved substantial improvements.

Among all tested approaches, the ensemble model provided the highest accuracy, achieving:

- **MAE: 2.16°C**
- **RMSE: 3.00°C**
- **R²: 0.887**

The findings demonstrate that combining multiple machine learning algorithms can improve weather forecasting accuracy and provide a more reliable approach for predicting temperature variations.

PM	Accelerator	Mission
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This project supports the PM Accelerator mission by applying data analytics and machine learning techniques to solve a real-world weather forecasting problem, demonstrating practical skills in data cleaning, exploratory data analysis, predictive modelling, and project documentation.

